

SavePredictedExams — SME

Economics

Higher level

Paper 3

SME Mark Scheme — Paper 3 Moderate

1. Use the question-specific mark scheme together with the mark bands. Award up to the maximum marks as indicated.
2. Whenever relevant, carry-over (OFR) marks must be awarded. If a candidate makes an error in calculation but then uses the incorrect figure appropriately and accurately in later question parts, the candidate may be fully rewarded.
3. A candidate may be penalised once for rounding errors, missing units, or not giving answers to 2 d.p. across all subparts.

Question 1 — Palm oil externalities

1. (a) (i) Define the term **externality**. [2]

Level		Marks
0	<i>The work does not meet a standard described by the descriptors below.</i>	0
1	<i>Vague definition.</i> The idea that an externality is a side-effect of an economic activity that affects a third party.	1
2	<i>Accurate definition.</i> An understanding that an externality is a cost or benefit imposed on a third party outside of a market transaction, that is not reflected in market prices.	2

(ii) Identify the external cost per tonne. [1]

Answer: External cost at Q_e = vertical gap between MSC and MPC at 60 m tonnes = **\$500 per tonne**.

Award [1] for the correct numerical answer with correct units.

(iii) Calculate the welfare loss arising from the overproduction of palm oil. [2]

Welfare loss triangle = $\frac{1}{2} \times \text{base} \times \text{height}$
 = $\frac{1}{2} \times (60 - 45) \text{ million} \times \500
 = $\frac{1}{2} \times 15 \times \$500 \text{ million} = \mathbf{\$3,750 \text{ million}}$ (or \$3.75 billion).

Any valid working [1]. Correct answer with units [1]. OFR applies.

(iv) Explain, using a diagram, why negative externalities of production lead to allocative inefficiency. [4]

Level		Marks
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0	<i>The work does not meet a standard described by the descriptors below.</i>	0
1	<i>There is a correct diagram OR an accurate written response.</i> For a diagram with $MSC > MPC$ and a welfare-loss triangle identified OR an explanation that at the free-market quantity $MSC > MSB$ [1], so too much is produced at too low a price, creating a welfare loss [1].	1-2
2	<i>There is a correct diagram AND an accurate written response.</i> For a diagram with $MSC > MPC$ showing the welfare-loss triangle AND an explanation that at Q_e $MSC > MSB$ [1] leading to allocative inefficiency / welfare loss [1]. Candidates who incorrectly label diagrams can be awarded a maximum of [3].	3-4

(v) State **one** characteristic of a common-pool resource. [1]

Any **one** of: rivalrous in consumption; non-excludable; subject to overuse/depletion.

(vi) Outline the meaning of the term **tragedy of the commons**. [2]

Marks	Level descriptor
0	<i>Does not meet the standard below.</i>
1	<i>Vague outline.</i> The idea that common resources tend to be overused.
2	<i>Accurate outline.</i> The tragedy of the commons is where common-pool (rivalrous but non-excludable) resources are depleted because each user acting in self-interest over-consumes them, imposing negative externalities on others.

(vii) Calculate the expected tax revenue. [2]

Tax revenue = tax per unit $\times$ quantity = $\$500 \times 45$ million = **\$22,500 million** (or \$22.5 billion).

Valid working [1]. Correct answer with units [1].

(viii) Explain, using a diagram, how a Pigouvian tax can internalise the negative externality. [4]

Level		Marks
0	<i>The work does not meet a standard described by the descriptors below.</i>	0
1	<i>There is a correct diagram OR an accurate written response.</i> For an externalities diagram showing MPC shifting up to $MPC + \text{tax}$ so that $MPC + \text{tax} = MSC$ OR an explanation that a tax equal to the external cost raises private costs to social costs [1], reducing output to Q_o and eliminating welfare loss [1].	1-2
2	<i>There is a correct diagram AND an accurate written response.</i> For a diagram showing MPC shifting up to coincide with MSC AND a clear explanation that the tax internalises the external cost [1] so the socially optimal quantity is produced, eliminating allocative inefficiency [1].	3-4

(ix) State whether palm oil is price elastic or price inelastic. [2]

Marks	Level descriptor
0	<i>Does not meet the standard below.</i>

1	States price inelastic without a reason, OR states a valid reason without the classification.
2	States that palm oil is price inelastic because the PED (−0.45) is between 0 and −1 in absolute value, i.e. $ \text{PED} < 1$, so a given percentage change in price leads to a smaller percentage change in quantity demanded.

(b) Recommend a policy that the Indonesian government could introduce to reduce the environmental damage caused by palm-oil production. **[10]**

Possible policies may include (not exhaustive):

- Imposition of a Pigouvian (specific) tax on palm-oil production
- Tradable pollution (carbon) permits
- Direct regulation and legislation (e.g. deforestation bans, protected zones)
- Subsidies for sustainable / certified RSPO production
- Government provision of information / eco-labelling / nudges
- International cooperation or trade-based certification requirements

Assessment Criteria (Paper 3, 10 marks)

Marks	Level descriptor
0	<i>The work does not meet the standard described below.</i>
1–2	Little understanding of the demands of the question; economic theory stated but not applied; no synthesis; no use of text/data.
3–5	Some understanding; relevant theory described; superficial synthesis/evaluation; limited use of text/data.
6–8	Understanding demonstrated; relevant theory explained; relevant diagram(s) included; appropriate synthesis/evaluation that is mostly balanced; use of text/data generally appropriate and relevant.
9–10	Specific demands addressed; theory fully explained; relevant diagram(s) included and fully explained; effective and balanced synthesis/evaluation; appropriate and relevant use of text/data to formulate a reasoned argument.

Sample response (abridged): A Pigouvian tax of \$500/tonne set equal to the external cost internalises the negative externality, shifting MPC to coincide with MSC. This reduces quantity from 60m to the socially optimal 45m tonnes, eliminating welfare loss of \$3.75 bn and raising \$22.5 bn of revenue for reinvestment in reforestation. However, Indonesia's palm-oil sector employs 16 million people — a tax may reduce employment and export earnings, and enforcement may be difficult in remote regions. The tax may therefore be combined with subsidies for sustainable producers. On balance, a Pigouvian tax is justified provided revenues are hypothecated to environmental restoration and transitional support for workers.

Question 2 — Ethiopia macroeconomic data

2. (a) (i) Define the term **real GDP**. [2]

Level		Marks
0	<i>The work does not meet a standard described by the descriptors below.</i>	0
1	<i>Vague definition.</i> The idea that real GDP measures a country's output in an economy.	1
2	<i>Accurate definition.</i> An understanding that real GDP is the total market value of all final goods and services produced in an economy in a given period, adjusted for inflation using a base-year price level.	2

(ii) Calculate Ethiopia's real GDP in 2023. [2]

Real GDP = (Nominal GDP / GDP deflator) × 100
= (155.80 / 140.3) × 100 = **\$111.05 billion** (to 2 d.p.).

Valid working [1]. Correct answer with units to 2 d.p. [1].

(iii) Calculate Ethiopia's real GDP per capita in 2023. [2]

Real GDP per capita = Real GDP / Population
= \$111.05 bn / 126.5 million = **\$877.86 per person** (OFR applies).

Valid working [1]. Correct answer with units [1].

(iv) State **two** limitations of real GDP per capita as a measure of living standards. [2]

Any **two** of (1 mark each):

- Does not reveal income distribution / inequality.
- Ignores non-market activities (household work, informal economy).
- Does not account for quality-of-life indicators (health, education).
- Does not reflect environmental degradation / sustainability.
- Averages can hide regional disparities.

(v) Calculate the rate of inflation between 2022 and 2023. [2]

Inflation rate = ((140.3 – 115.0) / 115.0) × 100 = **22.00%**.

Valid working [1]. Correct answer to 2 d.p. with % sign [1].

(vi) Outline **one** cost of high inflation to an economy. [2]

Marks	Level descriptor
0	<i>Does not meet the standard below.</i>
1	<i>Vague outline.</i> States one cost without explanation (e.g. "reduces purchasing power").
2	<i>Accurate outline.</i> Outlines a cost with explanation — e.g. loss of purchasing power for those on fixed incomes; menu costs; shoe-leather costs; reduced international competitiveness; uncertainty damaging investment; redistribution from savers to borrowers.

(vii) Explain, using an AD/AS diagram, how a rise in the world price of oil could lead to cost-push inflation in Ethiopia. **[4]**

Level		Marks
0	<i>The work does not meet a standard described by the descriptors below.</i>	0
1	<i>There is a correct diagram OR an accurate written response.</i> For an AD/AS diagram with SRAS shifting to the left OR an explanation that higher oil prices raise costs of production [1], reducing SRAS and pushing up the price level [1].	1-2
2	<i>There is a correct diagram AND an accurate written response.</i> For an AD/AS diagram with SRAS shifting to the left showing a rise in the price level and a fall in real output AND a clear explanation that oil is a key input to production [1], so higher oil prices raise firms' costs, decrease SRAS and cause cost-push inflation [1].	3-4

(viii) Calculate Ethiopia's unemployment rate in 2023. **[2]**

Unemployment rate = (Unemployed / Labour force) × 100 = (4.86 / 54) × 100 = **9.00%**.

Valid working [1]. Correct answer with % sign [1].

(ix) Distinguish between **cyclical** and **structural** unemployment. **[2]**

Marks	Level descriptor
0	<i>Does not meet the standard below.</i>
1	The distinction is expressed vaguely — e.g. identifies both without explaining the difference.
2	The distinction is clearly expressed — cyclical (demand-deficient) unemployment arises from a fall in AD during a recession, while structural unemployment arises from a mismatch between the skills of workers and the requirements of available jobs due to long-run changes in the structure of the economy.

(b) Recommend a policy that the Ethiopian government could introduce to reduce unemployment. **[10]**

Possible policies may include:

- Expansionary fiscal policy (increase G / cut T)
- Expansionary monetary policy (lower interest rates)
- Supply-side policies — investment in education / vocational training
- Labour-market reforms / reducing labour rigidities
- Infrastructure investment programmes to boost productive potential
- Targeted employment subsidies / apprenticeship schemes for youth

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6–8	Understanding demonstrated; relevant theory explained; relevant diagram(s) included; appropriate synthesis/evaluation that is mostly balanced; use of text/data generally appropriate and relevant.
9–10	Specific demands addressed; theory fully explained; relevant diagram(s) included and fully explained; effective and balanced synthesis/evaluation; appropriate and relevant use of text/data to formulate a reasoned argument.

Sample response (abridged): *Given that Ethiopia's unemployment is at 9.00% and much of it appears structural (youth, skills mismatch), supply-side policies such as investment in vocational training and infrastructure are likely to be most effective. Training shifts the LRAS curve to the right and reduces the natural rate of unemployment in the long run. However, supply-side policies are slow-acting and expensive; with inflation at 22% contractionary monetary policy may be needed, which would work against short-run employment goals. A balanced approach — combining targeted apprenticeship subsidies with fiscal prudence — is recommended.*